B.2 SUPERGLUE

This dataset can be found in the Huggingface hub under the name super_glue (Wang et al., 2019). We use the validation split of the subset named *boolq* (Clark et al., 2019), containing 3270 passages for answering Yes/No questions. We formulated prompts by combining the fixed instruction "Read the following passage and answer the question:", followed by the question field in the dataset example, and on a new line we write the passage field of the example. The plots below show these prompts' perplexity, token length frequency distribution, and scatter plot.

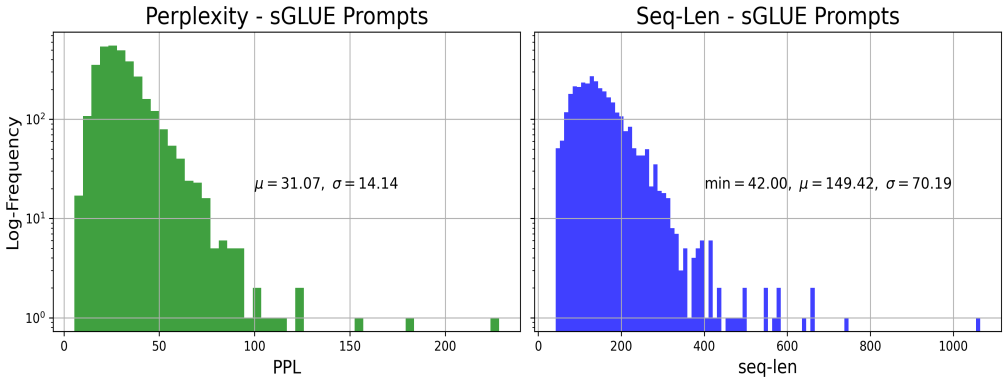

Figure 14: Log-frequency distributions for perplexity and sequence-length

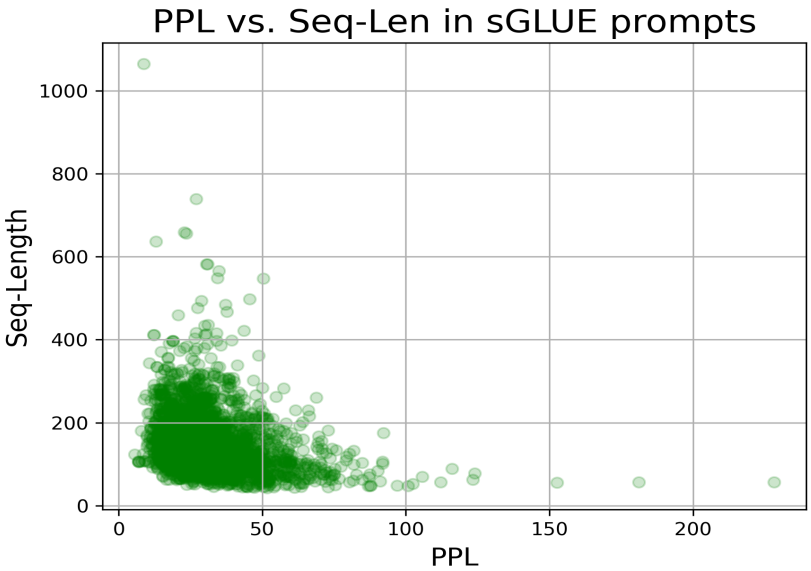

Figure 15: Scatter-plot showing perplexity vs. sequence-length

B.3 SQuAD-v2

The Stanford Question-Answering Dataset is a well-known span-based question-answering dataset that can be found in the Huggingface hub under the name squad_v2 (Rajpurkar et al., 2016). We use the validation split containing 11873 examples. We formulated prompts by combining three fields from each example, the title, the context and the question using the following form: We start with an instruction "Given a context passage from a document titled [title field goes here], followed by a question, try to answer the question with a span of words from the context:". Then after a new line the prompt continues with "The context follows:" followed by the context field, and then after another new line "The question is:" followed by the question field.

The plots below show these prompts' perplexity, token length frequency distribution, and scatter plot.

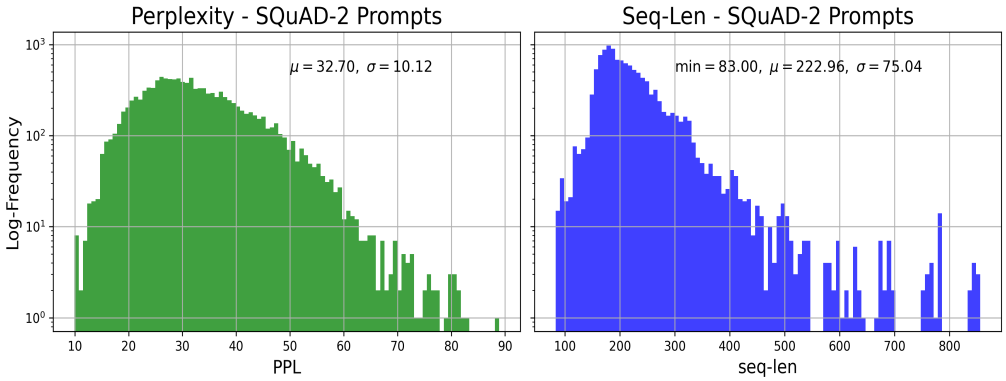

Figure 16: Log-frequency distributions for perplexity and sequence-length

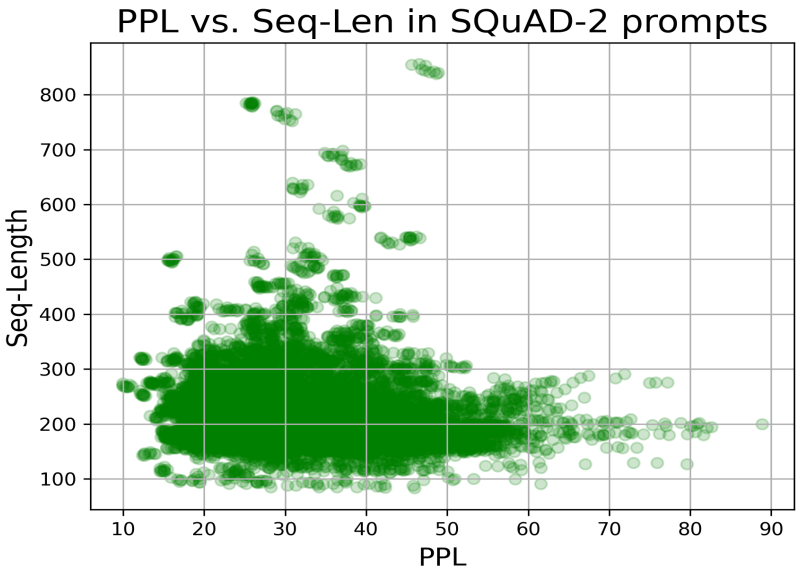

Figure 17: Scatter-plot showing perplexity vs. sequence-length

B.4 OPEN PLATYPUS

The Open-Platypus Dataset is associated with the Platypus project. We use the Huggingface dataset garage-bAInd/Open-Platypus containing 24926 prompts with instructions from the Platypus dataset's training split, as they appear, without any additional prefix or suffix. This dataset is focused on improving LLM logical reasoning skills and was used to train the Platypus2 models (Lee et al., 2023; Touvron et al., 2023; Hu et al., 2022). It is a filtered collection from multiple scientific, reasoning, and Q&A datasets including: Yu et al. (2020); Chen et al. (2023); Mihaylov et al. (2018); Sawada et al. (2023); Lee et al. (2023).

The plots below show these prompts' perplexity, token length frequency distribution, and scatter plot.

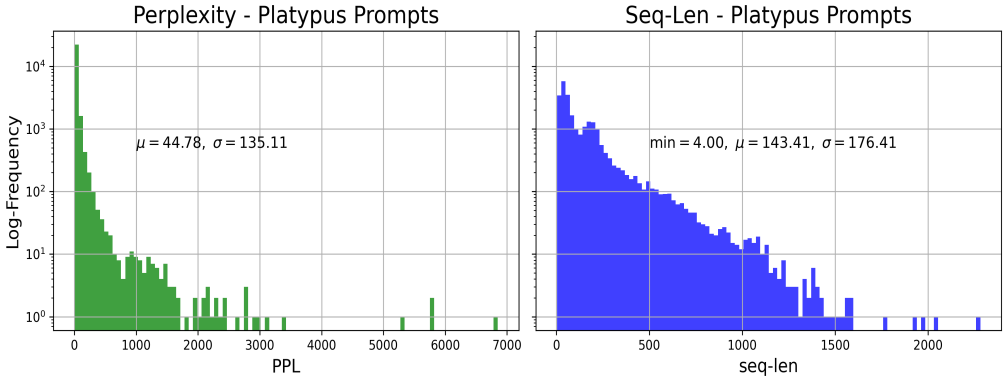

Figure 18: Log-frequency distributions for perplexity and sequence-length

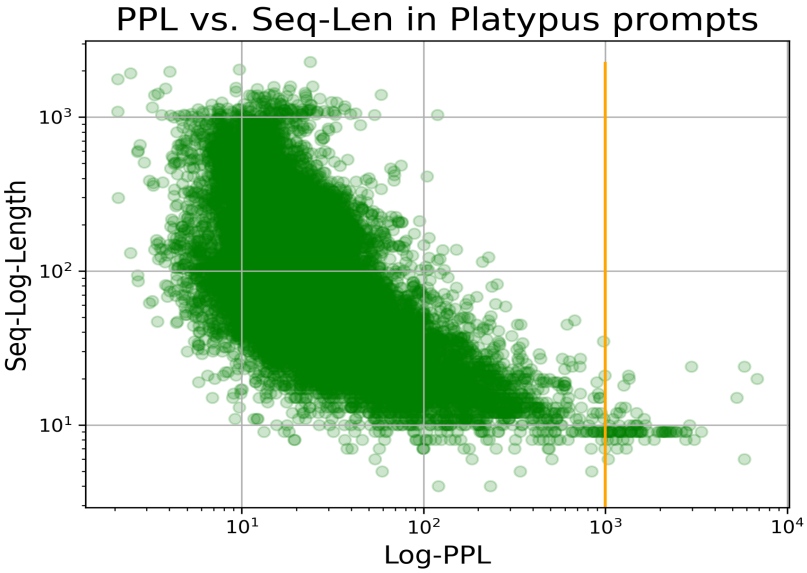

Figure 19: Scatter-plot showing perplexity vs. sequence-length

The shortest prompt is *Hello, AI!*.